



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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AgriVision Project

A CAPSTONE PROJECT REPORT

A Capstone Project Report submitted in partial fulfilment of the requirements for the Course of

ITA0507 - COMPUTER VISION

to the award of the degree of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE OF ENGINEERING –

ARTIFICIAL INTELLIGENCE

Submitted by

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SIMATS ENGINEERING

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Department of Computer Science and Engineering

ASSIGNMENT QUESTION

Course Code & Name:	ITA05 & Computer Vision
Assignment Title:	AgriVision – Drone-Based Crop Health Monitoring: Traditional vs. Deep-Learning Approach
Bloom’s Taxonomy Level:	L3 – Apply; L4 – Analyse; L5 – Evaluate; L6 – Create

Course Outcome(s) – CO:

CO2: Apply image enhancement, filtering, segmentation, and feature extraction techniques to preprocess and analyze images.

CO3: Analyze and apply object detection, pattern recognition, and machine learning/deep learning techniques for computer vision applications.

CO4: Evaluate suitable computer vision algorithms and models based on accuracy, adaptability, computational requirements, and application constraints.

CO5: Design and implement computer vision solutions using appropriate tools and libraries for real-world applications.

SDG Mapping (SDG 1–17):

SDG 2 – Zero Hunger

SDG 9 – Industry, Innovation and Infrastructure

SDG 15 – Life on Land

Problem Statement:

An agri-tech startup is planning to deploy “AgriVision”, a drone-imagery-based crop health monitoring system, for mid-sized farms. The system must process aerial images captured periodically across the growing season to detect early signs of crop disease, identify weed infestation, and estimate overall field health/yield potential, with accuracy sufficient to guide targeted spraying and reduce pesticide overuse. Drone images vary in illumination (time of day, cloud cover), altitude/scale, and can be affected by motion blur and sensor noise, making reliable analysis challenging.

The engineering team is considering two approaches:

- Traditional Computer Vision Methods: Colour-space and texture-based segmentation to isolate leaf/canopy regions, followed by handcrafted feature extraction (colour indices, texture descriptors, edge/shape features) and rule-based or classical-classifier disease/weed identification — fast, low-power, and interpretable, but less adaptive to varied disease symptoms, crop types, and field conditions.
- Deep Learning Methods: CNN-based disease-classification models and YOLO-style detectors trained on labelled crop-disease/weed datasets — capable of higher accuracy and better generalization across crop varieties and field conditions, but requiring substantial labelled training data, computational resources for training, and careful validation before field deployment.

As a computer vision engineer, apply image enhancement, segmentation, and feature detection/matching techniques to preprocess the drone imagery; compare the traditional computer vision and deep learning approaches for crop-disease detection, weed identification, and field-health estimation; and evaluate which approach is most suitable given the farm's computational resources (edge device on the drone vs. cloud

processing) and accuracy requirements. Analyze the trade-off between detection accuracy, adaptability to different crops/conditions, computational requirements, training-data availability, and real-world deployment constraints (battery life, connectivity, cost). Justify the selected approach based on performance versus adaptability, and design an end-to-end OpenCV-based solution explaining how it supports precision agriculture and sustainable farming practices.

Assessment Rubrics

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Problem Understanding & Vision-System Requirements (CO2)	10	Clearly identifies the drone-imagery crop-monitoring problem, accuracy/coverage requirements, and the role of enhancement, feature-based, and OpenCV-driven vision techniques in the solution.	Identifies most requirements with minor gaps and demonstrates good understanding of the problem.	Basic understanding of the problem is demonstrated, but some requirements are unclear.	Problem requirements are misunderstood or incorrectly interpreted.
Image Preprocessing, Enhancement, Segmentation & Feature Extraction (CO2)	10	Appropriately proposes preprocessing such as noise removal, illumination/haze correction, spatial/frequency-domain filtering, segmentation of leaf/canopy regions, and relevant feature extraction (edges, colour/texture, keypoints) for disease and weed analysis.	Appropriate preprocessing and feature-extraction techniques are selected with minor gaps.	Basic preprocessing techniques are identified but their application is limited.	Preprocessing/feature-extraction techniques are inappropriate or missing.
Comparison of Traditional CV and Deep-Learning Approaches (CO3/CO4)	15	Provides a comprehensive comparison of classical (colour/texture thresholding, edge/region-based segmentation, template/feature matching) and deep-learning (CNN-based disease classification, YOLO-based weed/pest detection) approaches based on accuracy, adaptability to field	Provides a relevant comparison covering most important factors.	Provides a basic comparison with limited analysis of performance and adaptability.	Comparison is incomplete, inaccurate, or lacks technical reasoning.

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
		conditions, computational cost, training requirements, robustness, and deployment suitability.			
Solution Design & Model/Algorithm Selection (CO4/CO5)	15	Designs a suitable end-to-end OpenCV pipeline (image enhancement → segmentation → feature/disease detection → yield/health estimate) and selects the most suitable approach with strong justification, clearly addressing training/validation, integration, and field deployment on drone/edge hardware.	Appropriate solution and models are selected with reasonable justification.	Solution is partially developed or model selection has limited justification.	No suitable solution design or model selection is provided.
Analysis, Evaluation & Justification (CO4)	15	Critically evaluates the accuracy–adaptability–computational-cost trade-off and convincingly justifies the selected approach using field-deployment, cost, and reliability considerations relevant to farmers/agronomists.	Provides good analysis and justification with minor limitations.	Provides general analysis but limited critical evaluation.	Little or no meaningful analysis or justification.
Implementation, Results & Visualization (CO5)	10	Successfully implements the proposed OpenCV/vision application; presents meaningful results, accuracy metrics, sample disease/weed detections, and visualizations (e.g., health maps, bounding boxes).	Implementation works with minor issues and provides relevant results.	Partial implementation with limited or inconsistent results.	Implementation is missing, non-functional, or results are not demonstrated.
Reflection, Societal/Industrial	5	Provides thoughtful reflection on design decisions,	Provides reasonable	Reflection is present but generic.	Reflection is missing or lacks meaningful

Criteria (CO Mapping)	Max Marks	Excellent	Good	Satisfactory	Needs Improvement
Relevance & SDG Mapping		challenges, learning outcomes, agricultural relevance, and clear SDG relevance.	reflection and SDG connection.		connection to the problem.

Suggested Conversion to 50 Marks

Problem Understanding & Vision-System Requirements	6
Image Preprocessing, Enhancement & Feature Analysis	6
Comparison of Traditional and Deep Learning Approaches	9
Solution Design & Model/Algorithm Selection	9
Analysis, Evaluation & Justification	9
Implementation, Results & Visualization	6
Reflection, Industrial/Societal Relevance & SDG Mapping	5
Total	50

Deliverables

1. Problem Analysis and Pseudocode – Include problem understanding, functional requirements, comparison of traditional vs. deep-learning approaches, selected approach, justification, and algorithm/pseudocode for the enhancement, segmentation, feature-detection, and disease/weed-recognition stages.
2. Implementation & Results – Include source code, dataset description (drone/field crop images, public crop-disease dataset, or synthetically generated samples), preprocessing, detection/classification, training/testing where applicable, evaluation metrics, visualizations (e.g., health maps, bounding boxes), sample detections, and comparative analysis.
3. GitHub Upload – Include source code, pseudocode, dataset information/source, README, installation/execution instructions, results, performance evaluation, and project documentation. GitHub repository link must be submitted with the assignment.

Assignment Learning Focus

The assignment enables students to apply, analyse, and evaluate computer vision techniques — image enhancement and segmentation, feature detection and matching, and object/pattern recognition — in a precision-agriculture scenario, while developing the ability to select and design an appropriate OpenCV-based solution based on accuracy, adaptability, computational requirements, and real-world field-deployment constraints.

1. Problem Statement and Problem Formulation



An agri-tech startup is planning to deploy AgriVision, a drone-imagery-based crop health monitoring system for mid-sized farms. The system must process aerial images captured periodically throughout the growing season to detect early signs of crop disease, identify weed infestation, and estimate overall field health and yield potential. The solution should provide sufficiently reliable information to guide targeted spraying and reduce unnecessary pesticide use.

Drone images can vary considerably because of illumination changes caused by time of day and cloud cover, changes in altitude and image scale, motion blur, and sensor noise. These variations make direct image analysis difficult. Therefore, image enhancement, filtering, segmentation, feature extraction, and recognition techniques are required before making field-level decisions.

The problem is formulated as a comparison between two computer vision approaches. The first is a traditional computer vision pipeline using colour and texture segmentation, handcrafted features, and rule-based or classical classifiers. The second is a deep-learning pipeline using CNN-based disease classification and YOLO-style object detection for weeds or pests. The final system must select or combine these approaches according to accuracy, adaptability, computational cost, training-data availability, battery life, connectivity, and deployment constraints.

2. Objective and Expected Outcomes

- Develop an end-to-end computer vision pipeline for drone-based crop health monitoring.
- Enhance drone images by reducing noise, correcting illumination variations, and improving image quality.
- Segment crop canopy, leaf, soil, and weed regions using suitable colour or region-based techniques.
- Extract colour, texture, edge, shape, and keypoint-based features for crop analysis.

- Compare traditional computer vision methods with CNN-based and YOLO-based deep-learning methods.
- Evaluate accuracy, adaptability, computational requirements, robustness, and deployment suitability.
- Generate crop-health indicators and visual outputs such as segmentation masks, detections, and health maps.
- Recommend a practical approach for edge-based drone processing and/or cloud-based processing.
- Support precision agriculture by enabling targeted intervention and reducing unnecessary pesticide use.

Expected Outcomes

- A functional OpenCV-based image preprocessing and segmentation pipeline.
- A comparative evaluation of traditional and deep-learning approaches.
- Disease and weed detection results with suitable evaluation metrics.
- Visualized crop-health information and sample detection outputs.
- A justified final solution considering real-world agricultural constraints.

3. Requirements, Constraints and Assumptions

3.1 Functional Requirements

- Input aerial images captured by a drone across different field locations and growth stages.
- Image enhancement and noise reduction.
- Crop/canopy and weed-region segmentation.
- Disease or abnormal-region identification.
- Field-health estimation and visualization.
- Performance evaluation using suitable test data and metrics.

3.2 Hardware and Software Requirements

- Drone or aerial-image acquisition platform.
- Edge computing device for optional onboard inference.
- Cloud or workstation GPU for model training and large-scale inference.
- Python programming environment.

- OpenCV for image processing and computer vision operations.
- NumPy and related Python libraries for numerical processing.
- Machine-learning/deep-learning framework such as TensorFlow or PyTorch, where applicable.

3.3 Constraints

- Limited drone battery capacity and onboard processing power.
- Variable illumination, altitude, scale, blur, and sensor noise.
- Limited availability of labelled drone crop-disease and weed images.
- Connectivity limitations when cloud processing is required.
- Training deep-learning models can require significant computational resources.
- False detections can lead to unnecessary spraying or missed disease symptoms.

3.4 Assumptions

- Drone images are captured with adequate spatial resolution for crop-level analysis.
- Training and test data represent the major field conditions considered during deployment.
- Ground-truth labels are available for supervised deep-learning experiments.
- A suitable threshold or classifier can distinguish healthy and abnormal crop regions for the traditional pipeline.
- The final health estimate is treated as a decision-support indicator rather than a guaranteed agronomic yield prediction.

4. Application of Relevant Course Knowledge / Concepts

- Image enhancement: contrast improvement, illumination correction, denoising, and filtering.
- Colour-space processing: RGB, HSV, or other colour representations for vegetation separation.
- Image segmentation: thresholding, region-based methods, morphology, and mask generation.
- Feature extraction: colour indices, texture descriptors, edges, contours, shape features, and keypoints.
- Feature detection and matching: useful for comparing repeated observations and identifying stable image regions.
- Pattern recognition: classification of healthy, diseased, and weed regions.

- Object detection: bounding-box detection of weeds, pests, or disease-related objects.
- Machine learning and deep learning: classical classifiers, CNNs, and YOLO-style detectors.
- Performance evaluation: accuracy, precision, recall, F1-score, IoU, inference time, and computational cost.
- OpenCV implementation: practical image acquisition, preprocessing, segmentation, visualization, and analysis.

5. Design / Proposed Solution / Methodology

The proposed AgriVision architecture follows an end-to-end pipeline: drone image acquisition → image quality assessment → enhancement and filtering → crop/weed segmentation → feature extraction or deep learning inference → disease/weed recognition → field-health estimation → visualization and reporting.

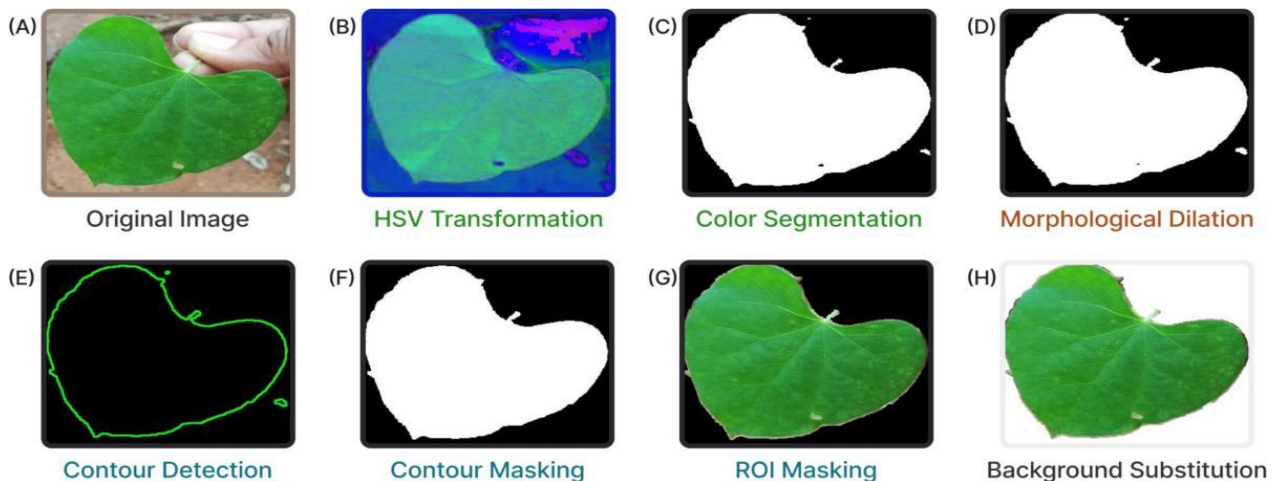
5.1 Image Acquisition and Quality Assessment

Drone images are collected periodically from the farm. Each image is checked for blur, exposure problems, noise, and insufficient resolution. Images that are severely degraded can be flagged for reacquisition or given lower confidence during analysis.

5.2 Image Enhancement

Noise reduction can be performed using Gaussian or median filtering. Contrast enhancement can be applied to improve visibility of crop regions. Illumination variations can be reduced using colour-space processing and local normalization. These operations improve the reliability of subsequent segmentation and feature extraction.

5.3 Segmentation



Vegetation can be separated from soil and background using colour thresholds in HSV or related colour spaces. Morphological opening and closing can remove small noise and fill gaps in the vegetation mask. Connected components or contours can then be used to identify candidate crop and weed regions.

5.4 Traditional Computer Vision Branch

The traditional branch extracts handcrafted features such as colour statistics, vegetation indices, texture descriptors, edges, contours, and shape information. These features can be passed to a rule-based system or a classical classifier. This branch is computationally light, interpretable, and suitable for resource-limited edge devices, but its performance can degrade when symptoms or field conditions differ from those used to design the rules.

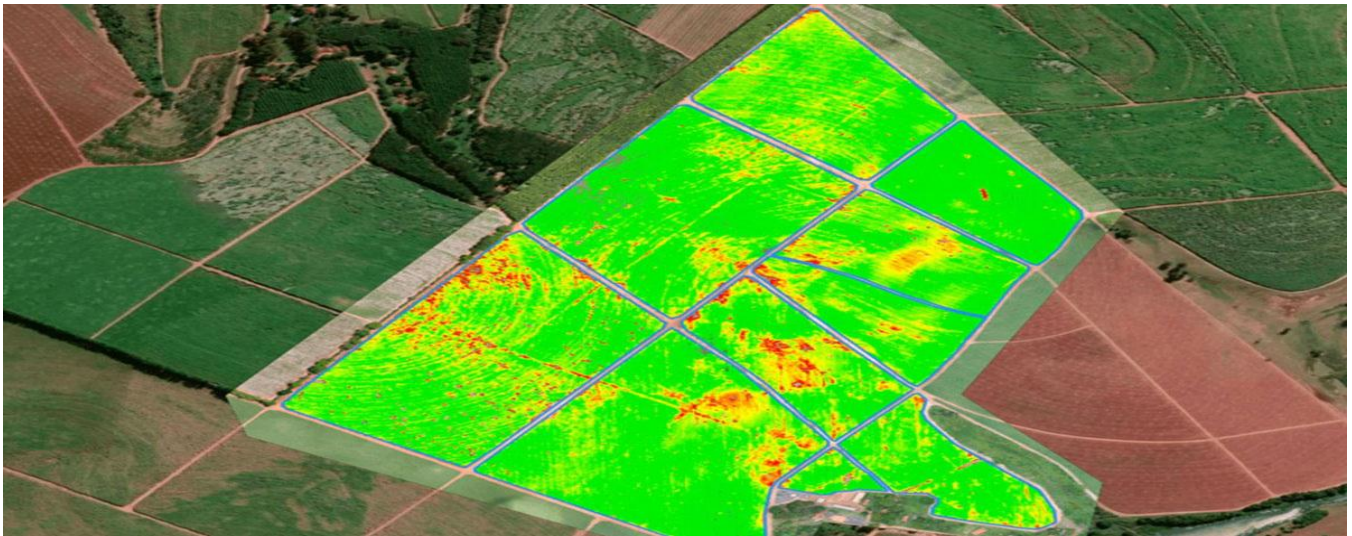
5.5 Deep Learning Branch



The deep-learning branch uses a CNN-based classifier for disease recognition and a YOLO-style detector for weed or pest localization. Training requires labelled images representing different crops, disease stages, illumination conditions, and field backgrounds. Validation data should be separated from training data to avoid overestimating performance.

5.6 Field-Health Estimation

A field-health score can be calculated from vegetation coverage, detected diseased area, weed density, and model confidence. The score can be mapped spatially to produce a simple health map. Areas with higher disease or weed probability can be prioritized for inspection or targeted treatment.



6. Algorithm / Pseudocode / Flowchart

6.1 Proposed Algorithm

1. Capture or load drone image.
2. Check image quality and resize if required.
3. Apply denoising and illumination/contrast enhancement.
4. Convert the image to a suitable colour space.
5. Segment vegetation/canopy regions.
6. Apply morphological operations to clean the segmentation mask.
7. Extract colour, texture, edge, contour, and shape features.
8. Run the traditional classifier/rules or deep-learning model.
9. Detect disease and weed regions and calculate confidence values.
10. Estimate field health from vegetation condition, disease area, and weed density.
11. Generate masks, bounding boxes, health indicators, and summary results.
12. Compare actual predictions with ground-truth labels and calculate performance metrics.
13. Store results for farm-level monitoring and decision support.

6.2 Pseudocode

BEGIN

Load drone image

Assess image quality

Resize image if required

Apply denoising and enhancement

Convert image to HSV/appropriate colour space

Segment vegetation and candidate weed regions

Apply morphological opening and closing

Extract colour, texture, edge and shape features

IF traditional approach is selected THEN

Apply rules/classical classifier

ELSE

Apply trained CNN/YOLO model

END IF

Calculate disease area and weed density

Estimate field-health score

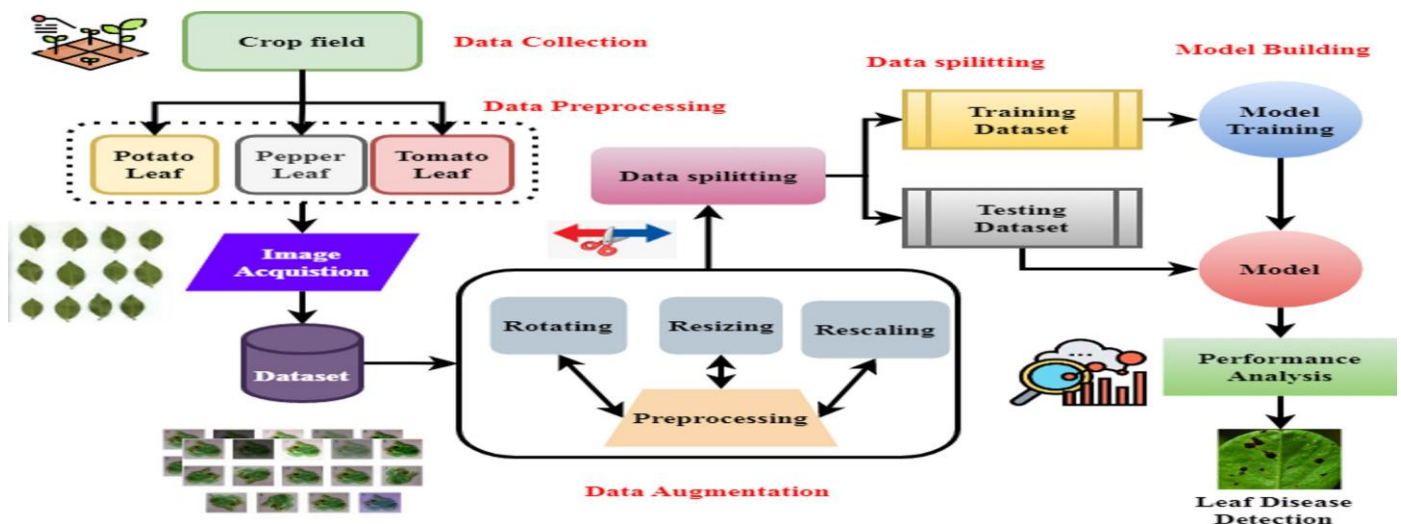
Generate visualization and report

Compare predictions with ground truth

Calculate accuracy, precision, recall, F1-score and IoU where applicable

END

6.3 Flowchart



7. Implementation / Source Code and Environment / Tools Used

The implementation can be developed in Python using OpenCV as the main computer vision library. NumPy can be used for matrix and image-array operations. For deep learning, a framework such as TensorFlow or PyTorch can be used to train and evaluate CNN/YOLO-style models.

- Programming Language: Python
- Image Processing Library: OpenCV
- Numerical Processing: NumPy
- Machine Learning/Deep Learning: suitable classical ML framework and TensorFlow/PyTorch where applicable
- Development Environment: Jupyter Notebook or Google Colab/workstation
- Dataset: drone/field crop images, public crop-disease images, or synthetically generated samples as permitted by the assignment

7.1 Representative OpenCV Implementation

The following implementation outline demonstrates the core preprocessing and vegetation segmentation stage. The actual disease/weed model can be integrated after this stage.

```
import cv2
import numpy as np
img = cv2.imread("crop_image.jpg")
# Resize
img = cv2.resize(img, (640, 480))
# Denoising
blur = cv2.GaussianBlur(img, (5, 5), 0)
# Convert to HSV
hsv = cv2.cvtColor(blur, cv2.COLOR_BGR2HSV)
# Vegetation threshold
lower = np.array([25, 40, 40])
upper = np.array([95, 255, 255])
mask = cv2.inRange(hsv, lower, upper)
# Morphological cleaning
kernel = np.ones((5, 5), np.uint8)
```

```

mask = cv2.morphologyEx(mask, cv2.MORPH_OPEN, kernel)
mask = cv2.morphologyEx(mask, cv2.MORPH_CLOSE, kernel)
# Vegetation percentage
vegetation_pixels = np.count_nonzero(mask)
total_pixels = mask.shape[0] * mask.shape[1]
vegetation_percentage = (vegetation_pixels / total_pixels) * 100
print("Vegetation Coverage:", vegetation_percentage, "%")
cv2.imwrite("vegetation_mask.jpg", mask)

```

8. Test Cases and Expected/Actual Results

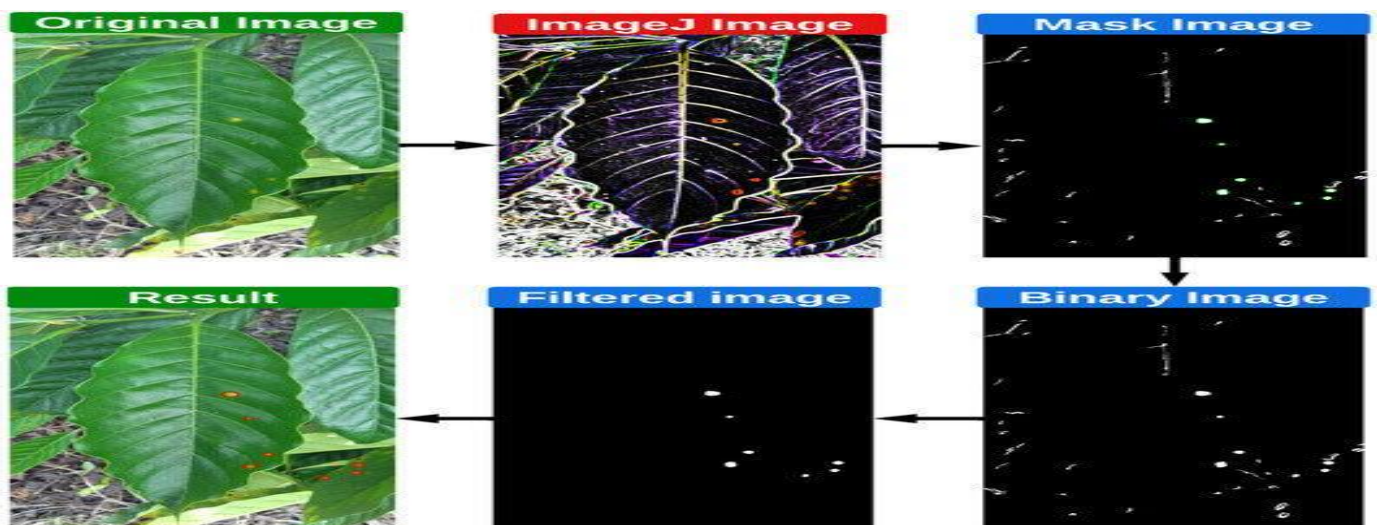
Test Case	Input Condition	Expected Result	Actual Result	Status
TC01	Normal daylight crop image	Clear vegetation segmentation	To be recorded during execution	Pending
TC02	Low-light/cloudy image	Enhanced image with usable segmentation	To be recorded during execution	Pending
TC03	Image with sensor noise	Noise reduced with preserved crop regions	To be recorded during execution	Pending
TC04	Image containing weeds	Weed regions detected/localized	To be recorded during execution	Pending
TC05	Image with visible disease symptoms	Diseased regions/class predicted	To be recorded during execution	Pending
TC06	Different image scale/altitude	Detection remains reasonably stable	To be recorded during execution	Pending

9. Execution Screenshots / Outputs



Insert the actual execution screenshots in this section after running the implementation. Recommended outputs include:

- Original drone image.
- Enhanced/denoised image.
- Vegetation segmentation mask.
- Disease-region visualization.
- Weed detection bounding boxes.
- Field-health map or heat map.
- Console output showing processing time and evaluation metrics.



10. Results and Validation

Results should be reported using the actual experimental measurements obtained from the selected dataset. The following table provides the recommended structure and should be populated after execution.

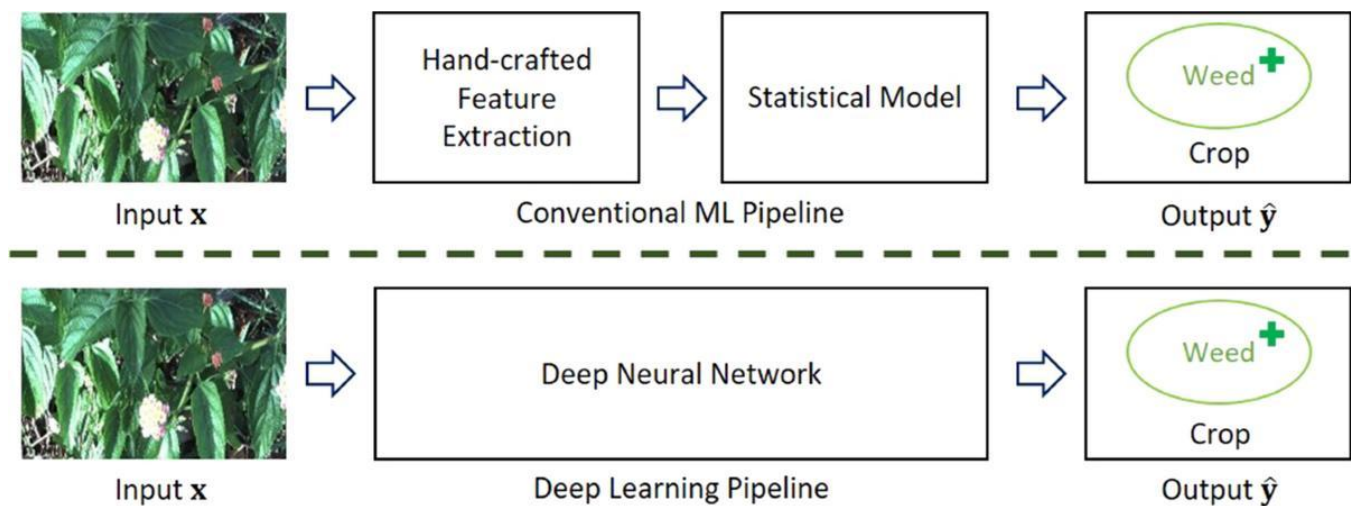
Metric	Traditional CV	Deep Learning	Observation
Accuracy	To be measured	To be measured	Compare classification correctness
Precision	To be measured	To be measured	Important for reducing false alarms
Recall	To be measured	To be measured	Important for detecting disease/weed cases
F1-score	To be measured	To be measured	Balances precision and recall
IoU	To be measured	To be measured	Useful for segmentation/detection overlap
Inference Time	To be measured	To be measured	Important for real-time/edge deployment

Validation should use separate test images that were not used to design thresholds or train the model. For classification, accuracy, precision, recall, and F1-score can be reported. For object detection or segmentation, IoU and related detection metrics can be included. In addition, inference time and resource usage should be measured to support deployment decisions.

11. Analysis, Comparison, Trade-offs and Justification

Criteria	Traditional CV	Deep Learning	Accuracy	Adaptability	Compute/Data	Deployment
Accuracy	Moderate; depends on features/rules	Generally higher with adequate training	DL advantage	DL advantage	Traditional advantage	Depends on hardware
Adaptability	Lower when conditions change	Better with representative training data	DL advantage	DL advantage	DL needs more data	Hybrid can help
Interpretability	High	Lower	Traditional advantage	Traditional advantage	Traditional advantage	Traditional advantage
Training Data	Low to moderate	High labelled-data requirement	Traditional advantage	DL requires diversity	Traditional advantage	DL training often cloud-based
Inference Cost	Low	Moderate to high	Traditional advantage	DL can be optimized	Traditional advantage	Edge favors lightweight models
Robustness	Sensitive to thresholds and lighting	Can learn complex patterns	DL advantage	DL advantage	Depends on validation	Hybrid/optimized DL suitable

Traditional computer vision is attractive for low-power edge processing because colour thresholding, morphological operations, handcrafted features, and simple classifiers are comparatively inexpensive. Its major limitation is reduced adaptability when crop varieties, disease symptoms, illumination, or background conditions change.



Deep learning is more suitable when high detection accuracy and generalization across varied field conditions are the priority. CNNs can learn complex disease patterns, while YOLO-style detectors can localize weeds and other objects. However, deep learning requires labelled training data, GPU resources during training, model validation, and potentially higher inference cost.

A practical final design is a hybrid or deployment-aware architecture. OpenCV preprocessing and segmentation can be performed before inference, while a lightweight trained deep-learning model can provide disease/weed recognition. If the drone has very limited compute, a lightweight traditional pipeline can provide rapid screening and upload selected images for cloud-based deep-learning analysis. This balances battery life, connectivity, accuracy, and cost.

12. Broader Considerations / SDG Relevance

12.1 SDG 2 – Zero Hunger

Early crop-disease and weed detection can help protect crop productivity and support more efficient agricultural practices.

12.2 SDG 9 – Industry, Innovation and Infrastructure

The project applies drone technology, computer vision, machine learning, and edge/cloud computing to develop an innovative agricultural monitoring system.

12.3 SDG 15 – Life on Land

Targeted spraying and improved field monitoring can reduce unnecessary chemical use and support more sustainable land management.

12.4 Responsible Deployment

- False positives and false negatives should be monitored because incorrect recommendations can affect farm cost and crop health.
- Models should be validated across crop varieties, seasons, lighting conditions, and geographical locations.
- Human/agronomist verification should be retained for high-impact treatment decisions.
- Data collection and storage should follow appropriate privacy and ownership practices.

13. Conclusion, Limitations and Possible Improvements

13.1 Conclusion

AgriVision demonstrates how computer vision can support precision agriculture through drone-based crop monitoring. The proposed pipeline combines image enhancement, segmentation, feature analysis, and recognition to identify disease and weed-related conditions and estimate field health. Traditional computer vision provides a fast and interpretable solution for constrained hardware, while deep learning offers stronger adaptability and recognition capability when sufficient labelled data and computational resources are available. A hybrid or deployment-aware strategy provides a practical balance between accuracy and real-world constraints.

13.2 Limitations

- Performance depends on image quality and representative training/testing data.
- Traditional thresholds may fail under severe illumination or crop-background changes.
- Deep-learning accuracy can fall when deployment conditions differ significantly from training data.
- High-quality labelled datasets for drone-based crop disease and weed detection may be limited.
- Actual field-level yield estimation requires additional agronomic and temporal information beyond a single image.

13.3 Possible Improvements

- Use lightweight optimized CNN/YOLO models for edge deployment.
- Increase dataset diversity across seasons, crops, locations, and illumination conditions.
- Apply data augmentation and systematic cross-validation.
- Integrate multispectral or hyperspectral information where available.

- Use temporal drone observations to track disease progression.
- Add GPS-based geospatial mapping for targeted field intervention.
- Develop confidence-based alerts and human verification before treatment recommendations.

14. Individual Contribution of Group Members

Member Name / Roll No.	Responsibility	Work Completed	Contribution %
Karthik Reddy	Dataset & Preprocessing	Collected/organized the crop image dataset, performed image enhancement, noise removal, and segmentation using OpenCV.	50%
Pavan Srinivaas	Model Development & Evaluation	Implemented disease/weed detection, compared traditional CV and deep-learning approaches, performed testing, analysis, and prepared results.	50%

15. References

1. OpenCV Documentation – Image Processing and Computer Vision functions.
2. Python Documentation – Python programming language and standard libraries.
3. NumPy Documentation – Numerical and array-processing operations.
4. TensorFlow or PyTorch Documentation – Deep-learning model development and evaluation.
5. YOLO Documentation/Research Literature – Real-time object detection methodology.
6. Public crop-disease and agricultural image datasets used for training/testing, with dataset names and source links to be added by the team.
7. Additional journal/conference papers used for crop disease detection, weed detection, image segmentation, and precision agriculture should be listed with complete bibliographic details.

One-Page Individual Reflection

Name: R S S N Pavan Srinivas

Roll No.: 192472360

Project Title: AgriVision – Drone-Based Crop Health Monitoring: Traditional vs. Deep-Learning Approach

Working on the AgriVision project helped me understand how computer vision techniques can be applied to real-world agricultural problems. The project focuses on analysing drone-based crop images to identify crop diseases, detect weeds, and estimate overall field health. Through this project, I gained practical knowledge of image preprocessing, enhancement, segmentation, feature extraction, and computer vision-based analysis.

My major contribution to the project was dataset preparation and image preprocessing. I worked on organizing the crop image dataset and understanding the different image conditions that can occur in drone-based agricultural monitoring. I also studied preprocessing techniques such as noise removal, filtering, colour-space conversion, thresholding, and morphological operations. These techniques helped me understand how image quality can affect the accuracy of subsequent crop and disease analysis.

A major learning outcome from this project was understanding the importance of image segmentation in agricultural computer vision. By separating vegetation from soil and background regions, the system can focus on the important crop areas. I also learned that drone images can have variations in illumination, altitude, scale, motion blur, and sensor noise. Therefore, suitable preprocessing and segmentation techniques are necessary to obtain reliable results.

I also learned about the differences between traditional computer vision and deep-learning approaches. Traditional methods use handcrafted features such as colour, texture, edges, and shape. They require less computational power and are easier to interpret. Deep-learning methods such as CNN-based classification and YOLO-style detection can learn complex patterns and provide better adaptability when sufficient labelled training data are available. Understanding these differences helped me appreciate the importance of selecting an appropriate method based on accuracy, computational resources, and deployment requirements.

During the project, I faced challenges in understanding how different environmental conditions affect crop-image analysis. Solving these challenges improved my analytical and problem-solving skills. I learned that a computer vision system should be tested under different conditions and evaluated using suitable performance measures such as accuracy, precision, recall, F1-score, and IoU.

The project also helped me understand the importance of precision agriculture and sustainable farming. Early detection of crop diseases and weeds can help farmers identify affected areas and apply treatments more selectively. This can reduce unnecessary pesticide use and support sustainable agricultural practices. The project is related to SDG 2 – Zero Hunger, SDG 9 – Industry, Innovation and Infrastructure, and SDG 15 – Life on Land.

Overall, the AgriVision project improved my knowledge of Python, OpenCV, image preprocessing, segmentation, computer vision concepts, and teamwork. It also helped me understand how academic computer vision techniques can be applied to practical agricultural problems. In the future, I would like to improve the system by using larger and more diverse datasets, advanced deep-learning models, multispectral images, GPS-based crop-health mapping, and real-time drone processing. This project provided valuable experience in developing a practical computer vision solution for precision agriculture.